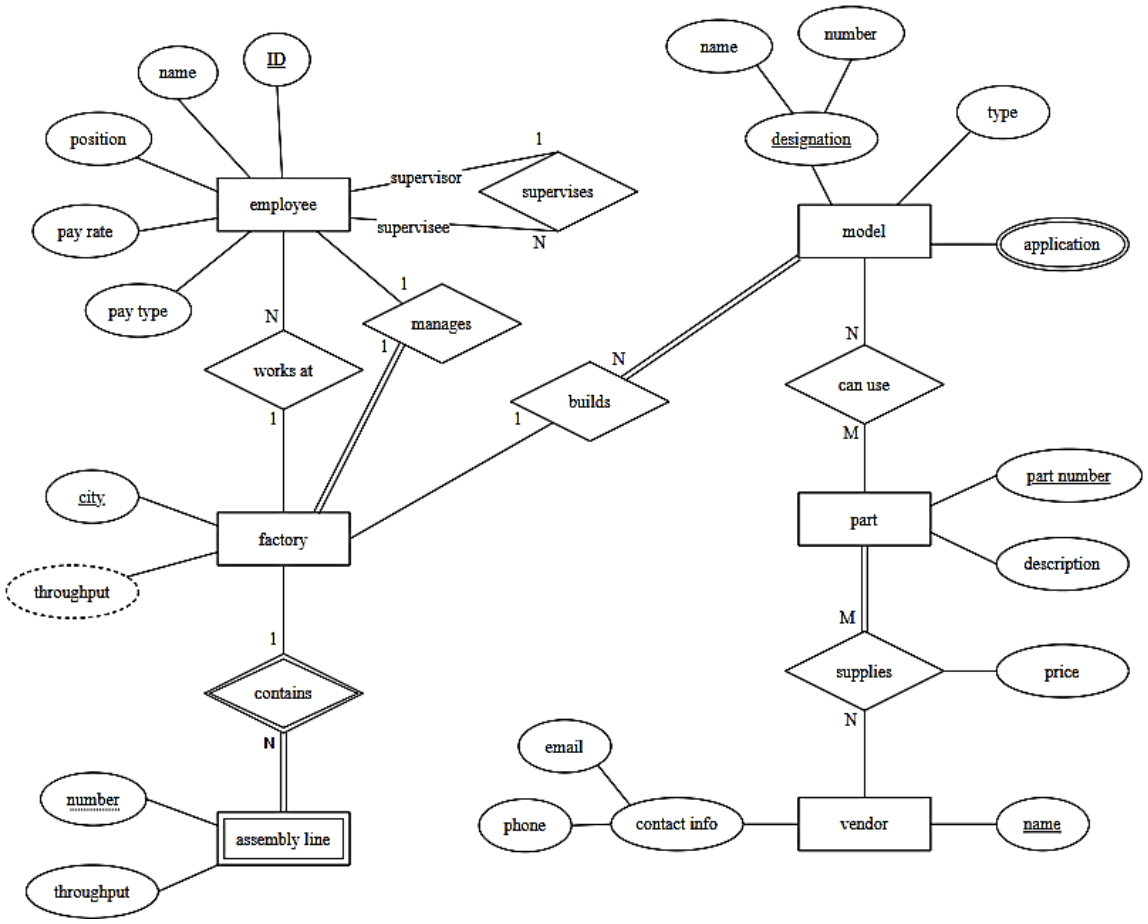
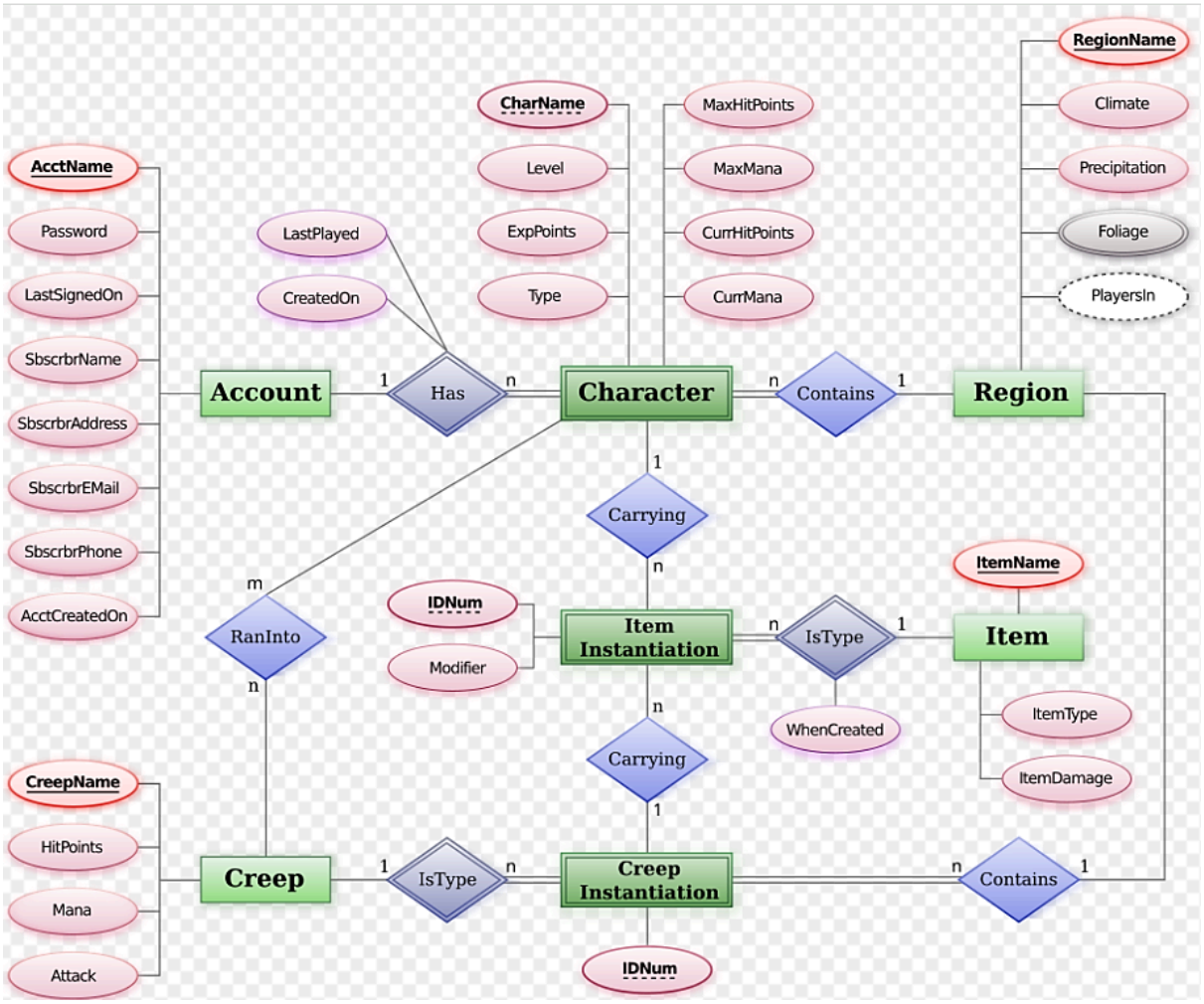
Assignment Questions in ER Model

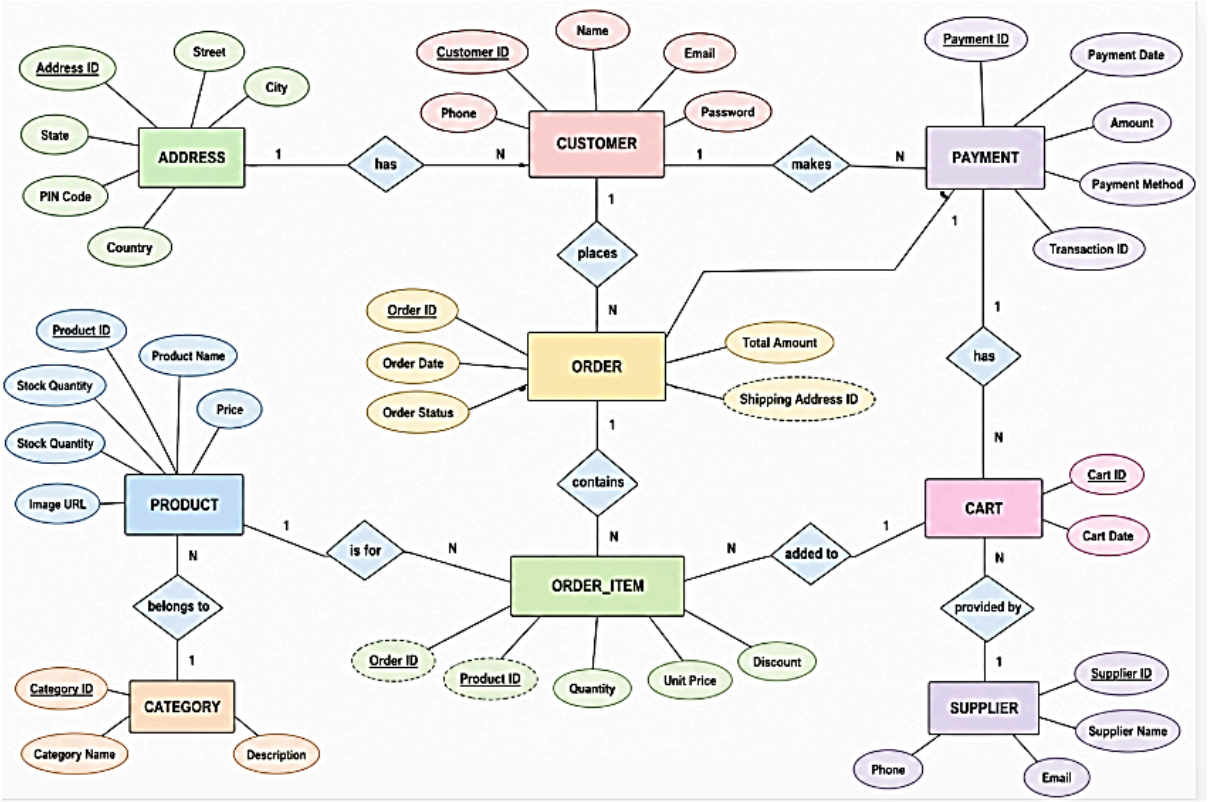
Design ER model for the following applications:

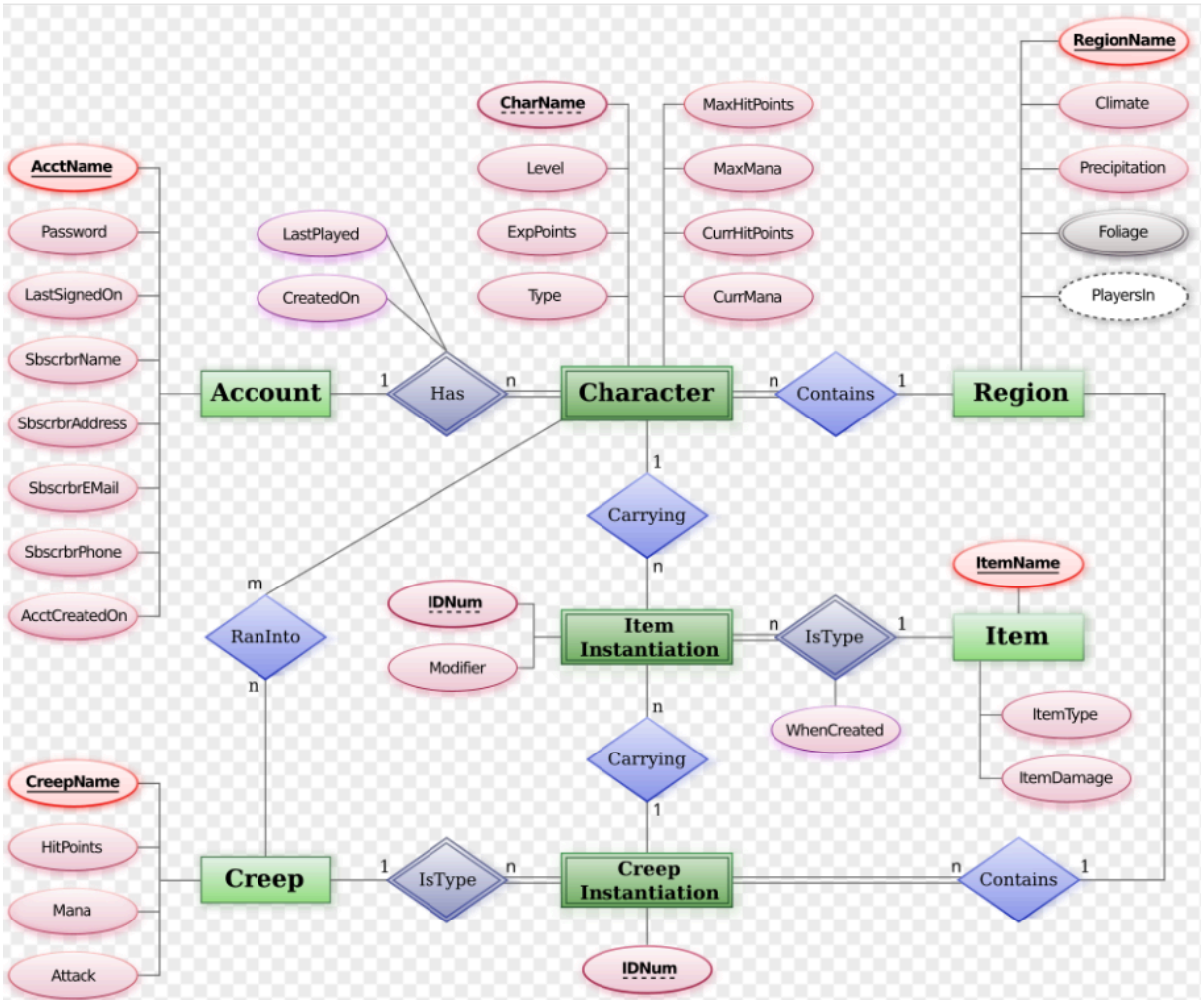
1. Library Management System:
   Construct an ER diagram for a library containing Books, Authors, Members, and Librarians. A member can borrow multiple books, and a book can be borrowed by different members over time.
2. Railway Reservation System:
   Construct an ER diagram for Passengers, Trains, Stations, Tickets, and Routes. A passenger can book multiple tickets, and a train can travel through multiple stations.
3. University Library System with Weak Entity:
   Design an ER diagram for a library where Books have multiple Copies. Each copy is identified using a copy number along with the Book ID. Include a weak entity and show its identifying relationship.
4. Construct an ER diagram that includes composite, multivalued, derived, and simple attributes for Online Food Delivery System.
5. A company wants to maintain an Employee Management System to store details of employees, departments, and projects. Each employee has an Employee ID, name, designation, salary, and contact details. Every employee works in one department, and each department can have many employees. Each department is managed by one employee. The company has several projects, and employees can work on one or more projects. Each project can have many employees. The system also stores the details of employees' dependents. Based on this scenario, draw an ER diagram showing the entities, attributes, relationships, primary keys, and cardinalities.
6. Design an ER diagram for a university database that includes generalization, specialization, weak entities, multivalued attributes, and M:N relationships.

Convert the following ER model into relational database table:




RegionName
Climate
Precipitation
Foliage
PlayersIn
CharName
MaxHitPoints
Level
MaxMana
ExpPoints
CurrHitPoints
Type
CurrMana
AcctName
Password
LastSignedOn
SbscrbrName
SbscrbrAddress
SbscrbrEMail
SbscrbrPhone
AcctCreatedOn
LastPlayed
CreatedOn
Account
1
Has
n
Character
n
Contains
1
Region
1
Carrying
n
m
RanInto
n
IDNum
Modifier
Item Instantiation
n
IsType
1
Item
ItemName
WhenCreated
ItemType
ItemDamage
n
Carrying
1
CreepName
HitPoints
Mana
Attack
Creep
1
IsType
n
Creep Instantiation
n
Contains
1
IDNum


ADDRESS
Address ID
Street
City
State
PIN Code
Country
has
CUSTOMER
Customer ID
Name
Email
Phone
Password
makes
PAYMENT
Payment ID
Payment Date
Amount
Payment Method
Transaction ID
places
ORDER
Order ID
Order Date
Order Status
Total Amount
Shipping Address ID
contains
has
CART
Cart ID
Cart Date
PRODUCT
Product ID
Product Name
Stock Quantity
Price
Stock Quantity
Image URL
is for
ORDER_ITEM
Order ID
Product ID
Quantity
Unit Price
Discount
added to
belongs to
CATEGORY
Category ID
Category Name
Description
provided by
SUPPLIER
Supplier ID
Supplier Name
Phone
Email


RegionName
Climate
Precipitation
Foliage
PlayersIn
CharName
MaxHitPoints
Level
MaxMana
ExpPoints
CurrHitPoints
Type
CurrMana
AcctName
Password
LastSignedOn
SbscrbrName
SbscrbrAddress
SbscrbrEMail
SbscrbrPhone
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